Experiment No : 7

Title:Evaluation of postfix expression.

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Program:

```
#include <stdio.h>
#include <ctype.h>

int stack[100], top = -1;

void push(int value) {
   stack[++top] = value;
}

int pop() {
   return stack[top--];
}

int main() {
   char exp[100];
   int i, V1, V2, result;

   printf("Enter postfix expression: ");
   scanf("%99s", exp);

   for (i = 0; exp[i] != '\0'; i++) {
      if (isdigit((unsigned char)exp[i])) {
         push(exp[i] - '0');
      } else {
         if (top < 1) {
            printf("Invalid expression!\n");
            return 1;
```

```
            }

            V2 = pop();
            V1 = pop();

            switch (exp[i]) {
               case '+': result = V1 + V2; break;
               case '-': result = V1 - V2; break;
               case '*': result = V1 * V2; break;
               case '/':
                  if (V2 == 0) {
                     printf("Division by zero!\n");
                     return 1;
                  }
                  result = V1 / V2;
                  break;
               default:
                  printf("Invalid operator!\n");
                  return 1;
            }
            push(result);
         }
      }

      if (top != 0) {
         printf("Invalid expression!\n");
         return 1;
      }

      printf("Result = %d\n", pop());
      return 0;
   }
```

Output:

Output

```
Enter postfix expression: 23*5+
Result = 11
```